

Smart Fall-Attenuation Attire (v3.0)

Zero-Cost Reusable Hybrid Pneumatic Framework with Core 4-Node Sensor Fusion

Document Type: Final Engineering Specifications

Propulsion: Onboard Hybrid Pneumatic Accumulator

Target Zones: Collar, Hips, Knee Joints

1. Technical Evolution & Executive Summary

Version 3.0 implements a sustainable structural paradigm shift by replacing traditional single-use, high-cost compressed CO₂ cylinders with a fully reusable, zero-operating-cost ****Onboard Hybrid Pneumatic Accumulator****. Prior iterations utilizing consumable cartridges imposed logistical constraints, including replacement downtime and recurring material costs following any sensor deployment or false trigger.

The optimized design integrates a permanent, lightweight atmospheric reservoir pre-pressurized by an onboard micro-pump. By combining this self-charging pneumatic chassis with a streamlined 4-node sensor framework (Collar IMU + Waist IMU + Waist Ultrasonic + Ankle Ultrasonics), the architecture delivers near-instant multi-zone impact mitigation alongside completely automated self-rearming capabilities.

2. Hybrid Pneumatic Accumulator Architecture

The propulsion hardware is consolidated into a permanent, self-contained sub-assembly anchored securely to the lumbar track of the waistband, operating across three structural phases:

- **The Compression Module:** A miniature, low-power 12V DC micro-air compressor is integrated into an insulated side pocket. When the garment is docked on a wireless charger or nighttime bedside dock, the pump automatically compresses ambient air into the storage reservoir.
- **The Permanent Atmospheric Reservoir:** A permanent, ultra-lightweight carbon-fiber wrapped aluminum air tank (operating at a pre-charged threshold of 100 – 150 PSI) holds standard atmospheric air. This eliminates the thermodynamic freezing drop and chemical safety hazards associated with rapid CO₂ gas expansion.
- **The Automated Self-Reset Loop:** Following an intentional deployment or a manually cleared false trigger, an exhaust valve opens to bleed the bladders. The onboard microcontroller then initiates the micro-compressor to repressurize the central tank back to standby thresholds within 3 to 5 minutes using ambient room air.

3. 4-Node Sensor Fusion & Logic Execution

Data processing is optimized through a lean 4-point sensory matrix that tracks core body posture and linear ground distance while isolating limb-flail artifacts:

1. **Collar IMU Node:** Measures rapid upper torso pitch changes and rotational velocity shifts.
2. **Waist IMU Node:** Tracks absolute pelvic center-of-mass angular rotation.

3. **Waist Ultrasonic Probe:** A downward-facing transducer tracking true vertical torso clearance relative to the floor ($V_z = \Delta d / \Delta t$).
4. **Ankle Ultrasonic Probes:** Hem-embedded low-profile nodes providing instant post-trip spatial mapping confirmation.

The Deterministic Deployment Command

A deployment signal is sent to the high-flow electronic solenoid manifold if and only if the differential angular velocity between the Collar and Waist IMUs matches a synchronized torso collapse, and the Waist Ultrasonic confirms ground proximity has dropped below 0.4 meters at a velocity breaching 2.5 m/s.

4. Pneumatic Flow & Multi-Zone Inflation Mechanics

Because ambient compressed air expands dynamically without freezing, gas distribution is routed seamlessly through structural textile channels directly from the central lumbar accumulator:

Target Bladder Zone	Pneumatic Conduit Path	Inflation Target (ms)	Biomechanical Protection
Collar / Neck	Vertical spine-line flexible conduit	≤ 80 ms	Cervical spine stabilization; lateral shoulder cushioning.
Left & Right Hips	Direct short-port lateral manifold lines	≤ 60 ms	Greater trochanter shielding to prevent femoral neck fractures.
Bilateral Knee Joints	Heat-sealed TPU trousers seam conduits	≤ 80 ms	Patella and tibial plateau protection during forward trips.

5. Garment Integration & System Advantages

Integrating a permanent hybrid reservoir provides distinct advantages over alternative wearable systems:

- **Zero Operational Expenses:** Eliminates consumable components, ensuring the device remains fully operational over years of active use without recurring supply chain costs.
- **Thermal Safety Compliance:** Utilizing standard atmospheric air avoids the intense endothermic freezing reaction characteristic of compressed CO₂, protecting fragile skin layers from thermal burns during rapid inflation.
- **Ergonomic Balance:** Housing the permanent carbon-fiber tank, mini-pump, and electronics pouch uniformly inside a reinforced lumbar belt assembly minimizes upper-torso weight, yielding an unobtrusive, daily-wearable safety profile.